ORIGINAL RESEARCH

Same jump height, different jump: Principal component analysis of take-off and landing force–time curves reveals movement strategies that conventional force-plate metrics do not capture

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ABSTRACT
Force-plate software reduces each countermovement jump (CMJ) to a few discrete metrics, and athletes with similar values are treated as equivalent. This study examined whether jumps with similar conventional metrics share the same force–time curve, and how much curve-shape variation those metrics capture. In total, 805 CMJ trials from 212 male footballers (median age 14.1 years), recorded on force plates, were normalised to body weight; the take-off phase and the landing phase, which ended at the lowest point of the centre of mass, were each time-normalised and concatenated. Principal component analysis retained four components that explained 74.8% of curve variance, each located almost entirely in the landing phase. Among trials matched on jump height, component-score dispersion was similar to that of the full sample (standard-deviation ratios 0.89–1.17; whole curve 1.01), and matching on relative peak force had no effect. Four vendor metrics, age and body mass explained 2.7–13.1% of each component (0.3–11.8% under athlete-grouped cross-validation). The first component was most closely related to landing duration ($r = -0.75$). Between-athlete differences accounted for 10.0–29.6% of component variance. With a fixed 1-s landing window, a take-off component emerged that the metrics largely explained. Equal jump heights conceal different, largely trial-specific landing strategies.

KEYWORDS
countermovement jump; principal component analysis; landing; force–time curve; football

1. Introduction

The countermovement jump (CMJ) performed on force platforms has become one of the most widely used tests in team sports. It is quick, places little demand on the athlete and is sensitive to training and to neuromuscular fatigue (Claudino et al., 2017; Gathercole et al., 2015). Commercial systems now detect the phases of the jump automatically and return dozens of variables for every trial, among them jump height, peak forces, phase impulses and phase durations (McMahon et al., 2018). In routine practice, however, decisions rest on a handful of these values. Practitioners are advised to choose a small set of metrics that suits the question being asked, whether

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it concerns performance profiling, fatigue monitoring or return to sport (Bishop et al., 2023). Athletes with similar values on those metrics are then treated, implicitly, as having jumped in a similar way.

That assumption sits uneasily with what is known about movement. The same outcome can be achieved through different coordinative solutions, and variability in how a task is executed is a functional property of the neuromusculoskeletal system rather than mere noise (Bartlett et al., 2007; Davids et al., 2003; Preatoni et al., 2013). For the vertical jump in particular, no single kinetic or temporal variable fully accounts for performance (Dowling & Vamos, 1993). Athletes of different jumping ability differ in how force is distributed across the unweighting, braking and propulsion phases (Sole et al., 2018), force–time characteristics differ by sex and sport (Laffaye et al., 2014), and the way force is produced within individual sub-phases relates to performance (Harry et al., 2018). Performance metrics are increasingly distinguished from strategy variables, such as countermovement depth and phase durations (Harry et al., 2024). Analyses of whole force–, velocity– and power–time curves have described training-induced changes across the curve rather than at single points (Cormie et al., 2009), and alternative, strategy-based CMJ variables can detect fatigue when jump height does not (Gathercole et al., 2015).

Waveform methods offer a way to describe the whole curve without choosing variables in advance. Principal component analysis (PCA) of time-normalised waveforms, and its functional counterpart, decompose a set of curves into a small number of orthogonal modes of variation, each with a score for every trial (Jolliffe, 2002; Jolliffe & Cadima, 2016; Ramsay & Silverman, 2005). In biomechanics these methods have described gait waveforms associated with knee osteoarthritis (Deluzio & Astephen, 2007), knee kinematics in the vertical jump (Ryan et al., 2006), running kinematics in Achilles tendinopathy (Donoghue et al., 2008) and on-water rowing (Warmenhoven et al., 2019), and principled ways of interpreting the components have been proposed (Brandon et al., 2013). For vertical jumps, dimensionality-reduction techniques describe performance-related aspects of the curve better than discrete points do (Richter et al., 2014b), principal component waveform analysis identifies improvements in jump performance (Floría et al., 2019), and alignment of the curves on biomechanical events improves the information that can be extracted from them (Moudy et al., 2018; Richter et al., 2014a).

Two gaps remain. First, the landing that follows every CMJ is usually discarded, even though it is recorded on the same plates in the same trial. Landing kinetics are of independent interest: stiff landings with high vertical forces have been associated with anterior cruciate ligament injury risk in young female athletes (Leppänen et al., 2017), jumping and landing variables have some utility as predictors of injury (Pedley et al., 2020), and CMJ landing characteristics change with the effort that athletes perceive themselves to be making (Hite et al., 2025). Second, although waveform studies show that curves carry information beyond discrete variables, the extent to which the metrics reported in routine monitoring fail to capture curve shape has rarely been quantified directly, and never, to our knowledge, by asking whether jumps matched on those metrics have matched curves. Routinely collected data from youth football, in which large numbers of tests are performed under field conditions, provide an opportunity to answer this question on a realistic scale.

The present study therefore analysed the concatenated take-off and landing force–time curves of male footballers, with three aims: (i) to identify the principal modes of variation of the curve and to interpret them biomechanically; (ii) to test whether trials matched on jump height, or on relative concentric peak force, show less curve-shape

variation than the sample as a whole; and (iii) to quantify how much of each mode of variation is explained by four conventional vendor metrics together with age and body mass, using evaluation schemes that respect the repeated trials of each athlete. In addition, the partition of curve-shape variation between athletes, sessions and trials, and the sensitivity of the conclusions to the definition of the landing phase, were examined.

2. Materials and methods

2.1. Study design and overview

This was a retrospective, cross-sectional analysis of routinely collected force-plate data. The analysis followed the pipeline in Figure 1. Quality-control rules, phase definitions, the number of retained components and the matched-band definition were fixed before any result was examined; the analysis does not include any data-driven tuning of these choices.

2.2. Participants and data source

Data were countermovement jumps performed by male football players during routine testing at a high-performance centre in Thane, Mumbai, India [TODO: name or describe the centre or programme, the testing period, and the data-sharing agreement]. All CMJ trials of male football players available through the VALD ForceDecks external application programming interface were extracted (833 trials from 215 athletes in 333 sessions). Before analysis, athletes, tests and trials were replaced by pseudonymous codes; the analysis files contain no names, team names, dates of birth, test dates or vendor identifiers, and the key linking codes to athletes is held separately by the first author. Age at test was computed before de-identification and rounded to 0.1 year. Of the 212 analysed athletes, 164 (77%) were under 18 years of age at their first analysed test. [TODO: ethics committee, approval number, consent of athletes and parental or guardian consent for those under 18 years, or the waiver of consent for the secondary use of routine monitoring data, and a statement that the study conformed to the Declaration of Helsinki].

2.3. Instrumentation and test protocol

Trials were recorded on VALD ForceDecks dual force platforms (VALD Performance, Brisbane, Australia), which record the vertical ground reaction force under each foot. Total vertical force was the sum of the two plates. Most trials were sampled at 1000 Hz; 60 trials were recorded at 500 Hz, which does not affect the analysis because every phase was resampled to a common grid (Section 2.6). Each session contained one or more trials (median 2 per session), and athletes contributed a median of 1 session (maximum 6). [TODO: warm-up, instructions given to athletes (including arm position and landing instructions), number of trials requested, rest between trials, and the criteria used by the testers to accept a trial]. The vendor software identified movement onset and returned, for every trial, the four metrics used in this study: jump height (from flight time), concentric peak force, eccentric braking impulse and concentric impulse over the first 100 ms of the concentric phase, together with contraction time,

flight time, countermovement depth and peak landing force, which were used only for interpretation.

2.4. Event detection and phase definitions

Three events were taken from the vendor output: movement onset t_{on}, take-off $t_{\mathrm{to}} = t_{\mathrm{on}} + T_{\mathrm{c}}$, where T_{c} is the contraction time, and touch-down $t_{\mathrm{td}} = t_{\mathrm{to}} + T_{\mathrm{f}}$, where T_{f} is the flight time (Harry et al., 2020; McMahon et al., 2018; Owen et al., 2014). The take-off phase ran from t_{on} to t_{to}. In the main analysis the landing phase ran from touch-down to the lowest point of the centre of mass (COM), the instant at which the downward velocity acquired during the fall is first brought to zero, which marks the end of impact absorption. The vertical velocity of the COM was obtained by integrating the impulse–momentum relationship from touch-down (Linthorne, 2001; Street et al., 2001),

$$v(t) = -\frac{g\,T_{\mathrm{f}}}{2} + \int_{t_{\mathrm{td}}}^{t} \frac{F(s) - mg}{m}\,\mathrm{d}s, \qquad t_{\mathrm{lc}} = \min\{\, t > t_{\mathrm{td}} : v(t) \geq 0 \,\}, \tag{1}$$

where F is the total vertical force, m the body mass, $g = 9.81\,\mathrm{m\,s^{-2}}$ and the initial value is the touch-down velocity implied by a symmetrical flight. The landing phase in the main analysis was $[t_{\mathrm{td}}, t_{\mathrm{lc}}]$. Because no consensus definition of the end of the CMJ landing exists, a sensitivity analysis used a fixed window from touch-down to 1 s after touch-down, which is the end of the vendor's trial record.

2.5. Quality control

Five rules were applied in order (Table 1). Trials were excluded if the vendor body mass was missing or invalid (R1), because every curve is divided by body weight; if the mass estimated from quiet stance before the jump differed from the vendor body mass by more than 5% (R2), which indicates an incorrectly entered mass or an athlete who was not still; if the mean force recorded during flight exceeded 50 N (R3), which indicates contact with the plate during flight or an event-detection error; if any of the four metrics or age was missing (R4); or, in the main analysis only, if the lowest point of the COM could not be identified within the recorded landing (R5).

2.6. Curve construction

Force was expressed in multiples of body weight, $\tilde{F}(t) = F(t)/(mg)$, so that curves of athletes of different mass are directly comparable and a value of one corresponds to quiet standing. Each phase $[t_a, t_b]$ was then resampled by linear interpolation at 101 equally spaced instants,

$$\tilde{F}_k^{(p)} = \tilde{F}\big(t_a + \tfrac{k}{100}\,(t_b - t_a)\big), \qquad k = 0, 1, \ldots, 100, \tag{2}$$

which aligns take-off at the end of the first block and touch-down at the start of the second, a simple form of landmark registration (Moudy et al., 2018; Ramsay & Silverman, 2005; Richter et al., 2014a). Time-normalising the whole trial to a single scale was rejected because the same event would then fall at different percentages in

different trials, and timing differences would be confounded with differences in shape. The flight phase was omitted: during flight the force is zero, so this portion of the curve carries no information about force production except its duration, which determines flight-time jump height (Linthorne, 2001); retaining it would allow the analysis to recover a conventional metric from the length of a block. Each trial was therefore represented by the vector $\mathbf{x}_i \in \mathbb{R}^{202}$ formed by the take-off block followed by the landing block.

2.7. Principal component analysis

PCA was applied to the $n \times 202$ matrix of curves (Jolliffe, 2002; Jolliffe & Cadima, 2016). Because all points share the same unit, the analysis used the covariance matrix of the centred curves, without scaling,

$$\mathbf{S} = \frac{1}{n-1}\sum_{i=1}^{n}(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top, \qquad z_{ik} = \mathbf{v}_k^\top(\mathbf{x}_i - \bar{\mathbf{x}}), \tag{3}$$

where $\mathbf{v}_k$ is the kth loading vector (unit length), λ_k its eigenvalue and z_{ik} the score of trial i on component k; the proportion of variance explained by component k is $\lambda_k / \sum_j \lambda_j$. Four components were retained, as specified before the analysis. The sign of each loading vector was fixed so that its largest absolute element was positive. Components were interpreted in three ways: by reconstructing the curves $\bar{\mathbf{x}} \pm 2\sqrt{\lambda_k}\,\mathbf{v}_k$, which show the effect of a two-standard-deviation change in the score (Brandon et al., 2013; Warmenhoven et al., 2019); by the share of each component located in the landing phase, $\sum_{j\in\mathcal{L}} v_{jk}^2$, where $\mathcal{L}$ indexes the 101 landing points; and by Pearson correlations of the scores with interpretable descriptors (peak landing force relative to body weight, landing duration $t_{\mathrm{lc}} - t_{\mathrm{td}}$, contraction time, countermovement depth, the four vendor metrics, age and body mass).

2.8. Matched-band dispersion

To test whether equal conventional scores imply equal curves (aim ii), trials between the 45th and 55th percentiles of jump height were selected, and, separately, trials between the same percentiles of concentric peak force relative to body weight. For each component, the dispersion ratio

$$\rho_k = \frac{\mathrm{SD}(z_k \mid i \in \mathcal{B})}{\mathrm{SD}(z_k)} \tag{4}$$

compares the standard deviation of the score among trials in band $\mathcal{B}$ with that in the full sample. A ratio close to one indicates that fixing the metric does not make curves more alike; a ratio well below one indicates that the metric constrains curve shape. The same comparison was made for the whole curve as $\rho = \{\mathrm{tr}(\mathbf{S}_\mathcal{B})/\mathrm{tr}(\mathbf{S})\}^{1/2}$, where $\mathbf{S}_\mathcal{B}$ is the covariance matrix within the band. Because athletes contributed several trials, 95% confidence intervals for ρ_k were obtained from 2,000 bootstrap resamples of athletes, each resample including all trials of the selected athletes, with band limits held fixed (Efron & Tibshirani, 1993).

2.9. Variance explained by conventional metrics

For aim (iii), each component score was regressed by ordinary least squares on the four vendor metrics and on age and body mass,

$$z_{ik} = \beta_{0k} + \sum_{q=1}^{6} \beta_{qk}\, w_{iq} + \varepsilon_{ik}, \tag{5}$$

and the coefficient of determination R^2_k was taken as the share of the component explained; $1 - R^2_k$ is the share left unexplained. Because in-sample R^2 is optimistic and trials of the same athlete are not independent, three further evaluations were made. In athlete-grouped five-fold cross-validation, athletes rather than trials were allocated to folds (Roberts et al., 2017), and the out-of-sample coefficient was computed as $1 - \sum_i (z_{ik} - \hat{z}^{(-f)}_{ik})^2 / \sum_i (z_{ik} - \bar{z}_k)^2$, where $\hat{z}^{(-f)}_{ik}$ is the prediction from the model fitted without the fold containing trial i. The regressions were also refitted with one randomly selected trial per athlete and with athlete means of all variables.

2.10. Structure of repeated measurements

To establish whether component scores describe stable characteristics of athletes or vary from trial to trial, each score was decomposed with a three-level random-intercept model,

$$z_{ajt} = \mu + a_a + b_{aj} + e_{ajt}, \quad a_a \sim N(0, \sigma^2_{\mathrm{A}}),\ b_{aj} \sim N(0, \sigma^2_{\mathrm{S}}),\ e_{ajt} \sim N(0, \sigma^2_{\mathrm{T}}), \tag{6}$$

for trial t of session j of athlete a, fitted by restricted maximum likelihood (Pinheiro & Bates, 2000). The shares σ^2_{A}, σ^2_{S} and σ^2_{T} of the total variance were reported; the between-athlete share is an intraclass correlation coefficient for a single trial (Hopkins, 2000; Koo & Li, 2016).

2.11. Sensitivity analyses

The complete analysis was repeated with the 1-s landing window. The width of the jump-height band was varied (40th–60th percentile, and median ± 1 cm and ± 2 cm).

2.12. Software and reproducibility

All analyses were run by a single R script in R 4.5.3 (R Core Team, 2026) with the packages dplyr, tidyr, ggplot2 (Wickham, 2016), patchwork and nlme (Pinheiro & Bates, 2000); every random step (cross-validation folds, the one-trial selection and the bootstrap) used seed 2026. The script writes every table, figure and number reported in this article. A preliminary version of this analysis was summarised in a conference report; rerunning the script from the de-identified data reproduced all 42 values checked against that report. Statistical reporting followed recommendations for sports medicine and exercise science (Hopkins et al., 2009). [TODO: repository URL or DOI once public].

3. Results

3.1. Sample

Of 833 trials, 805 trials from 212 athletes in 327 sessions met all quality-control criteria (Table 1); the 1-s landing window retained 811 trials, because the lowest point of the COM could not be identified in 6 trials. Athletes were 14.1 years of age at their first analysed test (median; interquartile range 12.0–16.8; range 8.8–37.9) and contributed 3.8 trials on average (maximum 16). Mean jump height was 28.2 ± 8.4 cm, and peak landing force ranged from 2.19 to 14.88 times body weight (Table 2).

3.2. Modes of variation of the force–time curve (aim i)

The mean curve showed the expected unweighting, braking and propulsion pattern in the take-off phase and a large, variable force peak in the landing phase (Figure 2A). Point-wise dispersion was much greater during landing than during take-off. The first component explained 37.3% of the variance, and the four retained components together explained 74.8% (Figure 2B); ten components were needed to explain 93.8%.

All four retained components were located almost entirely in the landing phase, which held 99.3–99.9% of their squared loadings (Table 3). PC1 (37.3%) contrasted landings in which force rose to a large peak at about 40% of the landing phase and remained well above body weight until the lowest point of the COM with landings in which force peaked early and then declined (Figure 3). Its strongest correlate was landing duration ($r = -0.75$): high scores denote short, stiff landings in which the downward momentum is arrested quickly. PC1 was moderately related to peak landing force ($r = 0.39$) and essentially unrelated to contraction time ($r = -0.04$) and countermovement depth ($r = 0.01$). PC2 to PC4 (16.0%, 12.6% and 8.9%) described the timing and number of landing force peaks, contrasting single early peaks, single later peaks and double-peaked landings, with their largest loadings at 26%, 16% and 9% of the landing phase. Peak landing force correlated with every component ($r = 0.22$–0.45), whereas the correlations with contraction time and countermovement depth did not exceed 0.15 and 0.18 in absolute value, and those with the vendor metrics did not exceed 0.34.

3.3. Curve shape among trials matched on conventional metrics (aim ii)

The jump-height band contained 82 trials from 51 athletes with a jump height of 27.1 ± 0.65 cm, a standard deviation 13 times smaller than that of the full sample. Within this band, take-off curves were similar but landing curves differed widely (Figure 4A). Matching on jump height reduced the dispersion of PC1 only modestly (ratio 0.89, 95% CI 0.70–1.06), and the ratios for the other components ranged from 0.95–1.17; every confidence interval included one (Table 4, Figure 4B). The dispersion of the whole curve was unchanged (ratio 1.01). Matching on relative concentric peak force had no discernible effect (component ratios 1.00–1.05; whole curve 1.00). The result did not depend on the width of the band: across four band definitions the PC1 ratio ranged from 0.87–0.89, and all ratios from 0.84–1.17.

3.4. Variance explained by conventional metrics (aim iii)

The four vendor metrics, age and body mass explained 2.7–13.1% of the variance of the retained components, leaving 86.9–97.3% unexplained (Table 5, Figure 5). Explained variance was lower under athlete-grouped cross-validation (0.3–11.8%), and similar when one trial per athlete (4.3–17.8%) or athlete means (5.4–21.2%) were analysed, so the low values were not a consequence of treating repeated trials as independent.

Between-athlete differences accounted for 10.0–29.6% of the variance of the component scores, differences between sessions of the same athlete for 2.3–34.9%, and differences between trials within a session for 35.5–74.8% (Table 5). PC1 was the most athlete-specific component (29.6% between athletes, 34.9% between sessions); for PC2 to PC4, 61.6%, 74.8% and 69.6% of the variance arose between trials of the same session.

3.5. Sensitivity to the definition of the landing phase

With the landing phase ending 1 s after touch-down, the impact occupied only the first few per cent of the landing block and the remainder approximated quiet standing (Figure 6). The four components then explained 62.2% of the variance, and PC3 became a take-off component (96.3% of its squared loading in the take-off phase) that was strongly related to contraction time ($r = -0.79$) and relative concentric peak force ($r = 0.77$). This component was largely explained by the conventional metrics ($R^2 = 68.3\%$; 66.4% under cross-validation), whereas the three landing components remained poorly explained ($R^2 = 5.3$–19.1%; Table 6, Figure 5). PC1 was again a landing component related to peak landing force ($r = 0.71$) and landing duration ($r = -0.57$). Matching on jump height reduced PC1 dispersion somewhat more under this definition (ratio 0.82, 95% CI 0.69–0.96), but the dispersion of the whole curve was again unchanged (ratio 1.01).

4. Discussion and implications

4.1. Principal findings

This study asked whether countermovement jumps that score alike on conventional force-plate metrics also look alike, and how much of the shape of the force–time curve those metrics capture. Three findings stand out. First, when take-off and landing are analysed together, the dominant modes of variation lie in the landing, and the first describes how quickly the athlete arrests the fall. Second, jumps matched closely on jump height, or on relative peak force, were scarcely more alike in curve shape than jumps in general. Third, jump height, concentric peak force, eccentric braking impulse, early concentric impulse, age and body mass together explained no more than 13.1% of the variance of any landing component, and less under cross-validation. Equal scores on these metrics therefore do not imply an equal jump. A further finding qualifies the interpretation of the components as strategies: most of their variance arose within athletes, between sessions and between trials of the same session.

4.2. Comparison with previous work

The results extend earlier evidence that discrete force-plate variables summarise the jump incompletely. Dowling and Vamos (1993) found that no single kinetic or temporal variable explained vertical jump performance well, Richter et al. (2014b) showed that dimensionality-reduction techniques captured performance-related features of the jump better than discrete points, and principal component waveform analysis has been used to identify the features of the curve associated with improvements in jump performance (Floría et al., 2019). These studies, like most CMJ waveform analyses (Cormie et al., 2009; Harry et al., 2018; Sole et al., 2018), examined the propulsive part of the jump and asked how curve features relate to performance. The present study reverses the question and asks how much of the curve the performance metrics account for; the matched-band analysis gives a direct and intuitive answer, because it holds jump height almost constant and observes what remains.

The sensitivity analysis clarifies where the unexplained variation lies. With the 1-s landing window a take-off component emerged that described the duration and amplitude of the propulsive force and was explained to about two-thirds by the conventional metrics. This agrees with the close relationships between contraction time, relative peak force, impulse and jump height that follow from impulse–momentum mechanics (Linthorne, 2001; McMahon et al., 2018), and with the distinction now drawn between performance metrics and strategy variables (Harry et al., 2024). The metrics that practitioners usually report are therefore reasonable summaries of the take-off, but they are almost silent about the landing. This is not surprising, because the four vendor metrics describe propulsion, and the result should not be read as a failure of those metrics. Rather, it shows that the landing recorded in every trial contains substantial information that routine reports omit.

The importance of the landing is consistent with the wider literature. Landings that arrest the body quickly with high forces have been linked to injury risk in young female athletes (Leppänen et al., 2017), and landing kinetics have some utility as predictors of injury (Pedley et al., 2020). PC1, which separated short, stiff landings from longer, more compliant ones, corresponds closely to this construct. Its relation to landing duration follows from impulse–momentum: because the landing phase in the main analysis ends when the downward velocity reaches zero, the mean force over the phase is fixed by the touch-down velocity and the duration of the landing, so a shorter landing must be accompanied by higher forces. The weaker correlations with peak landing force indicate that the component describes the time course of force rather than its maximum alone.

4.3. Trial-to-trial variability and the interpretation of strategies

Between-athlete differences accounted for at most 29.6% of the variance of any component, which corresponds to poor single-trial reliability by conventional benchmarks (Koo & Li, 2016), and for PC2 to PC4 most of the variance arose between trials of the same session. The landing differences among jumps matched on jump height therefore reflect, to a large extent, how a given athlete landed on a given trial rather than a fixed characteristic of the athlete. This is in keeping with the view that movement variability is an intrinsic and often functional feature of skilled performance (Bartlett et al., 2007; Davids et al., 2003; Preatoni et al., 2013), and with the known sensitivity of CMJ variables to the number of trials and to protocol details (Cormack et al., 2008; Heishman et al., 2020). Landing behaviour is also likely to depend on instructions and

perceived effort (Hite et al., 2025), which were not standardised for landing in routine testing. The word strategy should therefore be understood here as the way in which a trial was executed; whether athletes have stable landing strategies can be judged only from repeated, standardised landings.

4.4. Methodological implications

Two analytical choices had a large influence on the results and should be reported in any similar study. The first is the definition of the end of the landing. Ending the landing at the lowest point of the COM confines the landing block to impact absorption and makes landing timing comparable across trials; a fixed window spends most of the block on quiet standing and compresses the impact into a few points, which lowered the variance captured by four components from 74.8% to 62.2% and changed the composition of the components. The second is the scaling of the curve. Covariance PCA of body-weight-normalised force gives weight to each part of the curve in proportion to its variance, and because landing forces are larger and more variable than take-off forces, the take-off contributes little to the leading components unless the landing is short or analysed separately. Neither choice is right or wrong, but they answer different questions, and the matched-band and regression results should be interpreted with them in mind. Alignment on further events, such as the landing force peak (Moudy et al., 2018), functional PCA (Ramsay & Silverman, 2005; Warmenhoven et al., 2019) and statistical parametric mapping (Pataky et al., 2013) are natural extensions.

4.5. Practical applications

For practitioners, the results have four implications. First, two athletes with the same jump height may land very differently, so ranking or grouping athletes on jump height alone conceals differences in landing mechanics. Second, the landing is recorded in every CMJ at no additional cost, and simple landing descriptors, in particular landing duration to the lowest point of the COM and peak landing force relative to body weight, capture much of the dominant landing component and could be added to routine reports. Third, because landing shape varies considerably between trials, decisions should be based on the mean of several trials, or on repeated sessions, rather than on a single landing. Fourth, unusual landing curves should prompt closer observation of how an athlete lands, under standardised instructions, rather than a diagnosis; their relationship with injury in footballers has not been established.

4.6. Strengths and limitations

The strengths of the study include a large set of routinely collected trials analysed in full, the inclusion of the landing phase, a direct test of the equivalence of matched jumps, evaluation schemes that respect the repeated structure of the data, explicit sensitivity analyses of the main analytical choices, and a single script that reproduces every reported result from the de-identified data.

Several limitations should be noted. The landing was not standardised in routine testing, so part of its variation may reflect instructions, habits or effort rather than neuromuscular capacity. The end of the landing depends on the body mass and on a velocity estimated from flight time, both of which carry error (Street et al., 2001), and

could not be found in 6 trials. The sample comprised male footballers from a single centre over a wide age range (8.8–37.9 years), most of them adolescents; maturity status was not recorded, although growth and maturation affect jump performance in youth athletes (Lloyd & Oliver, 2012; Malina et al., 2015), and the results may not generalise to female athletes, other sports or other age groups. The regressions were linear and used four metrics; other vendor variables, including landing variables, would explain more of the landing components, and non-linear models might explain somewhat more, although the weak correlations suggest that the gain would be limited. PCA on linearly time-normalised curves combines differences in timing and in amplitude. Finally, the number of components was fixed in advance rather than chosen from the data, which avoids tuning but means that the remaining components, which together explained about a quarter of the variance, were not interpreted.

4.7. Future work

Future studies should test whether landing components are stable when landings are standardised and repeated, relate them to injury, fatigue and return-to-play outcomes in prospective designs, and examine female athletes and other sports. Separate analysis of the take-off and landing, registration on additional events and functional or multivariate methods that combine curve shape with timing would help to separate strategy from execution variability. Establishing reference values for landing components across age and maturity would make the information usable in routine monitoring.

5. Conclusion

In male footballers, countermovement jumps matched closely on jump height, or on relative peak force, differed almost as much in the shape of their force–time curves as jumps in general. The dominant differences lay in how athletes absorbed the landing, above all in how quickly the fall was arrested, and the four conventional vendor metrics, together with age and body mass, explained little of them. These differences were, however, largely trial-specific. The landing phase of the CMJ, which is recorded in every trial but rarely reported, adds information that conventional take-off metrics do not provide, and it deserves a place in routine force-plate reports, provided that it is interpreted over repeated trials.

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Data availability statement

The de-identified trial data (curves and vendor metrics keyed by pseudonymous codes), the analysis script and all output tables are available in the project repository [TODO: URL or DOI]. The raw ForceDecks exports and the key linking codes to athletes are not shared [TODO: state the data owner and the terms under which raw data may be requested].

Declaration of generative AI use

[TODO: The authors must state whether and how generative AI was used. Suggested wording, to be confirmed: an AI assistant was used to help write and run the analysis code and to draft the manuscript text; the authors reviewed all code, results and text and take full responsibility for the content.]

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Table 1. Quality-control flow (main analysis).

Step	Trials	Sessions	Athletes
All male football CMJ trials	833	333	215
R1: valid body mass	828	329	212
R2: quiet-stance mass within 5R3: no force on plate during flight	811	327	212
R4: complete metrics and age	811	327	212
R5: end of landing (lowest COM) found	805	327	212

Rules R1–R4 were fixed before any result was examined; R5 applies only to the main landing definition (touch-down to the lowest point of the centre of mass), and the 1-s landing window therefore retained 811 trials. Sessions, test occasions on which an athlete performed one or more trials. COM, centre of mass.

Table 2. Characteristics of the analysed sample (805 trials, 212 athletes).

Variable	n	Mean ± SD	Median (IQR)	Range
Age at first test (years)	212	16.01 ± 5.94	14.10 (12.00 to 16.80)	8.80 to 37.90
Body mass at first test (kg)	212	53.31 ± 15.90	51.73 (40.86 to 64.03)	25.03 to 95.97
Jump height, flight time (cm)	805	28.19 ± 8.43	26.96 (21.62 to 34.69)	8.54 to 51.47
Concentric peak force (N)	805	1269.26 ± 420.74	1227.01 (923.47 to 1582.99)	473.52 to 3073.22
Concentric peak force (× BW)	805	2.35 ± 0.38	2.29 (2.11 to 2.51)	1.65 to 4.70
Eccentric braking impulse (N s)	805	27.80 ± 15.43	24.60 (16.59 to 35.70)	0.00 to 95.18
Concentric impulse, first 100 ms (N s)	805	62.24 ± 26.66	58.70 (41.76 to 79.14)	1.51 to 187.93
Peak landing force (× BW)	805	5.39 ± 1.97	4.97 (3.88 to 6.67)	2.19 to 14.88
Contraction time (s)	805	0.78 ± 0.19	0.77 (0.66 to 0.90)	0.31 to 1.48
Countermovement depth (cm)	805	−26.55 ± 9.90	−27.13 (−33.55 to −20.71)	−52.39 to 0.00
Touch-down to lowest COM (s)	805	0.25 ± 0.11	0.24 (0.15 to 0.31)	0.07 to 0.93
Trials per athlete	212	3.80 ± 2.28	3.00 (2.00 to 5.00)	1.00 to 16.00
Trials per session	327	2.46 ± 0.78	2.00 (2.00 to 3.00)	1.00 to 7.00
Sessions per athlete	212	1.54 ± 0.80	1.00 (1.00 to 2.00)	1.00 to 6.00

Age and body mass are per athlete at the first analysed trial; all other rows are per trial unless stated. Jump height, from flight time. Countermovement depth is the vendor estimate of centre-of-mass displacement (negative downwards). Touch-down to lowest COM, duration of the landing phase in the main analysis. BW, body weight; COM, centre of mass; IQR, interquartile range.

Table 3. Retained principal components of the concatenated take-off and landing curve (main analysis).

	Variance	Cumulative	Landing share	Largest	Pearson r with				
Component	(%)	(%)	of loading (%)	loading at	LPF	JH	CT	CMD	LD
PC1	37.3	37.3	99.7	Landing 40%	0.39	0.13	−0.04	0.01	−0.75
PC2	16.0	53.3	99.9	Landing 26%	0.22	0.13	0.02	0.00	−0.14
PC3	12.6	65.9	99.5	Landing 16%	0.39	0.34	0.00	−0.14	−0.32
PC4	8.9	74.8	99.3	Landing 9%	0.45	0.24	0.15	−0.18	−0.04

Landing share of loading, percentage of the squared loading vector that falls in the landing block (the remainder falls in the take-off block). Largest loading at, phase and percentage of the phase at which the loading has its largest absolute value. LPF, peak landing force (× BW); JH, jump height; CT, contraction time; CMD, countermovement depth; LD, landing duration (touch-down to lowest COM). Signs are fixed so that the largest absolute loading is positive.

Table 4. Dispersion of component scores in the full sample and among trials matched on jump height or on relative concentric peak force (main analysis).

	SD, full	Jump-height band		Relative peak-force band	
Component	sample	SD	Ratio (95% CI)	SD	Ratio (95% CI)
PC1	7.28	6.45	0.89 (0.70–1.06)	7.33	1.01 (0.77–1.16)
PC2	4.77	4.62	0.97 (0.77–1.16)	5.01	1.05 (0.82–1.24)
PC3	4.22	4.03	0.95 (0.74–1.13)	4.25	1.00 (0.81–1.17)
PC4	3.56	4.16	1.17 (0.81–1.44)	3.75	1.05 (0.87–1.23)
Whole curve[a]	11.92	11.99	1.01	11.95	1.00

Matched bands comprise trials between the 45th and 55th percentiles of jump height (26.0–28.2 cm; 82 trials, 51 athletes) or of concentric peak force relative to body weight (2.25–2.32 BW; 81 trials, 58 athletes). Ratio, SD in the band divided by SD in the full sample; 95% CI from 2,000 athlete-level bootstrap resamples. Component SDs depend on the number of grid points and are comparable only within this table; ratios are scale-free. [a]Square root of the summed point-wise variance of the 202-point curve.

Table 5. Variance of each component score explained by the four vendor metrics, age and body mass, and decomposition of component-score variance (main analysis).

	R^2 of component score on metrics, age and body mass (%)					Variance of component score (%)		
Component	In-sample	Adjusted	Athlete CV	One trial per athlete	Athlete means	Between athletes	Between sessions	Within session
PC1	7.7	7.0	4.0	10.0	11.1	29.6	34.9	35.5
PC2	2.7	2.0	0.3	4.3	5.4	21.4	17.0	61.6
PC3	13.1	12.4	11.8	17.8	21.2	22.9	2.3	74.8
PC4	9.1	8.4	6.3	11.1	13.8	10.0	20.3	69.6

Predictors: jump height, concentric peak force, eccentric braking impulse, concentric impulse over the first 100 ms, age and body mass (ordinary least squares). In-sample and adjusted, all 805 trials. Athlete CV, out-of-sample R^2 from five-fold cross-validation with athletes, not trials, allocated to folds. One trial per athlete, one randomly drawn trial per athlete (212 trials). Athlete means, scores and predictors averaged within athlete. Variance shares from a three-level random-intercept model (athlete, session within athlete, trial within session) fitted by restricted maximum likelihood.

Table 6. Main analysis compared with the sensitivity analysis in which the landing phase ended 1 s after touch-down.

Quantity	Main: landing to lowest COM	Sensitivity: 1-s landing
Trials / athletes after QC	805 / 212	811 / 212
Variance explained, PC1–PC4 (%)	37.3, 16.0, 12.6, 8.9	20.2, 16.6, 15.1, 10.2
Cumulative, PC1–PC4 (%)	74.8	62.2
Landing share of loading, PC1–PC4 (%)	99.7, 99.9, 99.5, 99.3	96.4, 92.5, 3.7, 80.2
PC1 SD ratio, jump-height band (95% CI)	0.89 (0.70–1.06)	0.82 (0.69–0.96)
Whole-curve SD ratio, jump-height band	1.01	1.01
In-sample R^2, PC1–PC4 (%)	7.7, 2.7, 13.1, 9.1	19.1, 5.3, 68.3, 17.8
Athlete-CV R^2, PC1–PC4 (%)	4.0, 0.3, 11.8, 6.3	13.4, 2.8, 66.4, 15.1
Between-athlete variance, PC1–PC4 (%)	29.6, 21.4, 22.9, 10.0	45.3, 18.7, 26.5, 31.6
r(PC1, landing peak force)	0.39	0.71
r(PC3, contraction time)	0.00	−0.79

COM, centre of mass; CI, confidence interval; CV, cross-validation. Between-athlete variance from the three-level random-intercept model. In the 1-s window PC3 is a take-off component (Figure 6), which explains its correlation with contraction time.

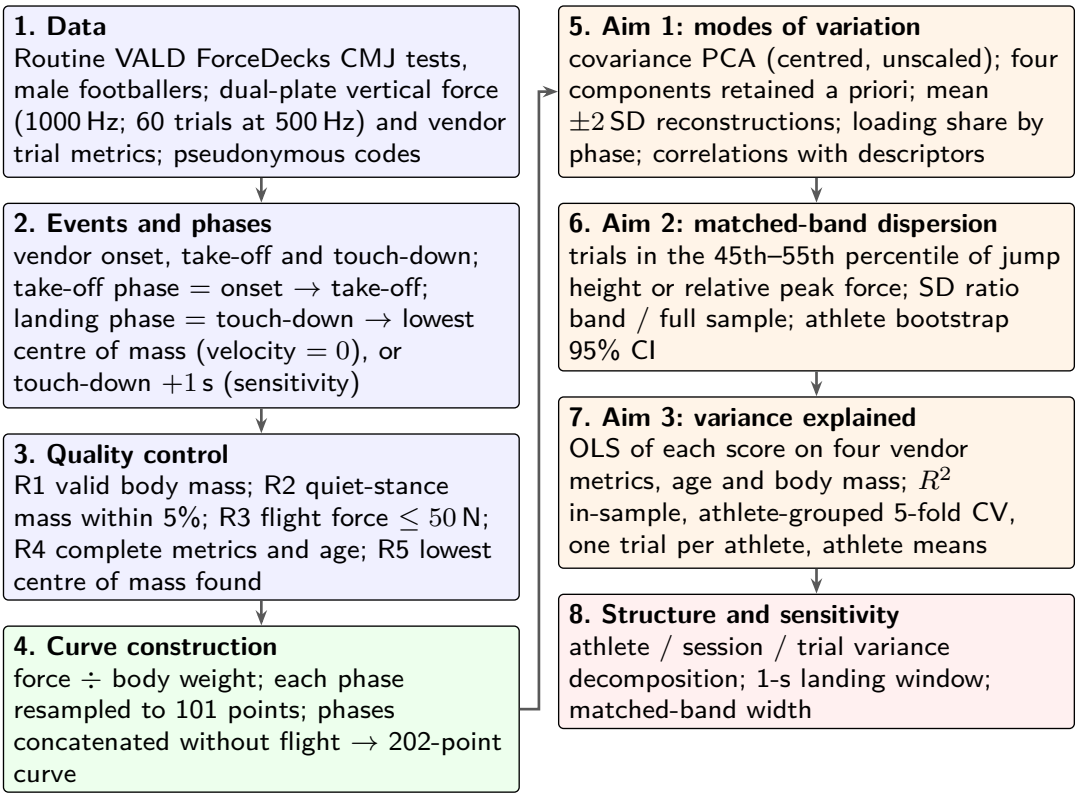


Figure 1. Analysis pipeline. Steps 1–4 build one 202-point, body-weight-normalised force–time curve per trial; steps 5–7 address the three aims; step 8 examines the structure of the repeated measurements and the sensitivity of the conclusions to analytical choices. All decision rules were fixed before the results were examined. CMJ, countermovement jump; PCA, principal component analysis; OLS, ordinary least squares; CV, cross-validation; CI, confidence interval.

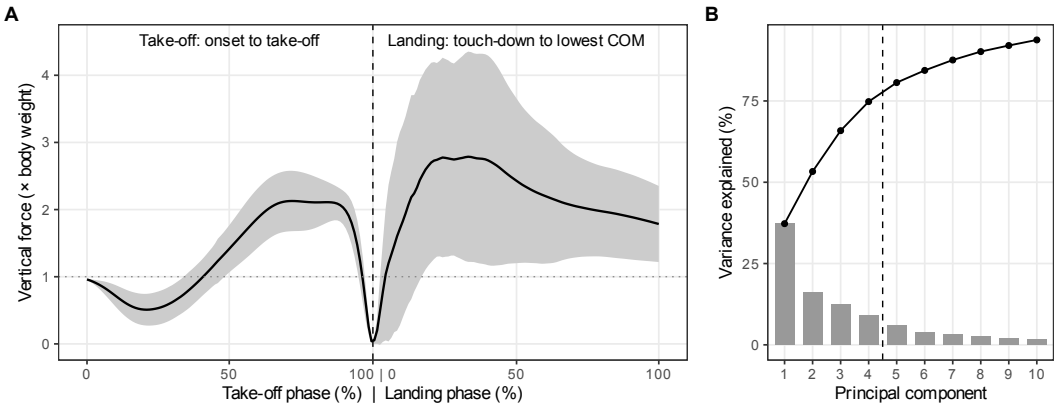


Figure 2. (A) Mean body-weight-normalised vertical force (line) ± 1 SD (shaded) of the 805 analysed countermovement jumps. The take-off phase (movement onset to take-off) and the landing phase (touch-down to the lowest point of the centre of mass, COM) were each time-normalised to 0–100% and concatenated; flight is omitted. The dotted line marks body weight. (B) Variance explained by each of the first ten principal components (bars) and cumulatively (line); the dashed line marks the four retained components.

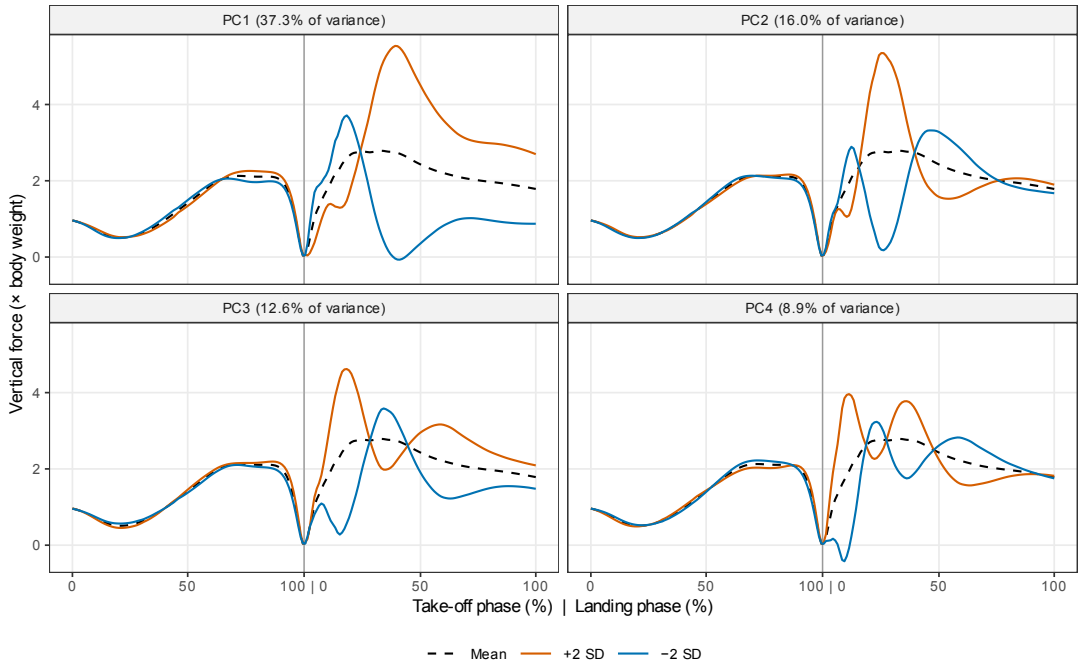


Figure 3. Interpretation of the four retained principal components (main analysis). Each panel shows the mean curve (dashed) and the mean curve plus (vermillion) or minus (blue) two standard deviations of the component score. Values below zero in the reconstructions are an artefact of adding a linear component to the mean and do not correspond to observed trials. The vertical line separates the take-off and landing phases.

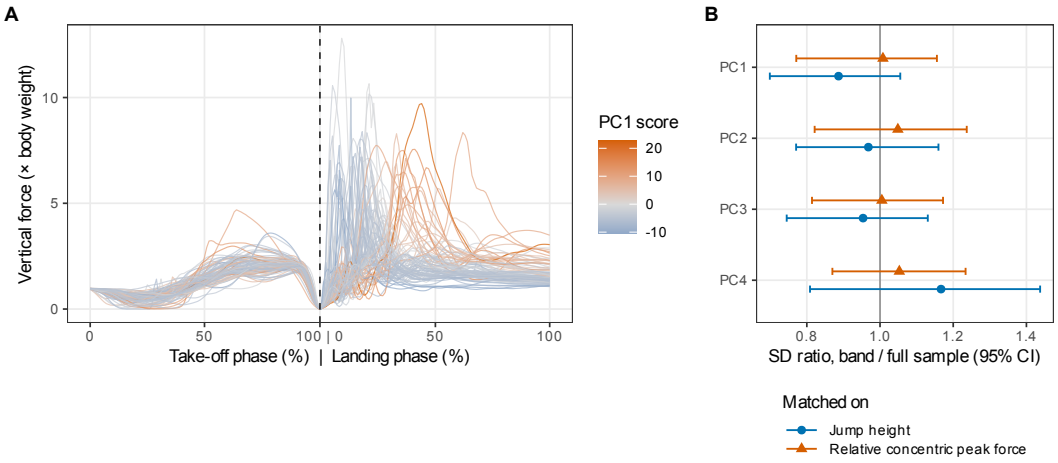


Figure 4. Trials matched on jump height do not share a common force–time curve. (A) Curves of the 82 trials between the 45th and 55th percentiles of jump height (27.1 ± 0.65 cm), coloured by their PC1 score. (B) Ratio of the standard deviation of each component score among trials matched on jump height (circles) or on relative concentric peak force (triangles) to that in the full sample, with athlete-level bootstrap 95% confidence intervals. A ratio of 1 indicates that matching on the metric leaves curve-shape variation unchanged.

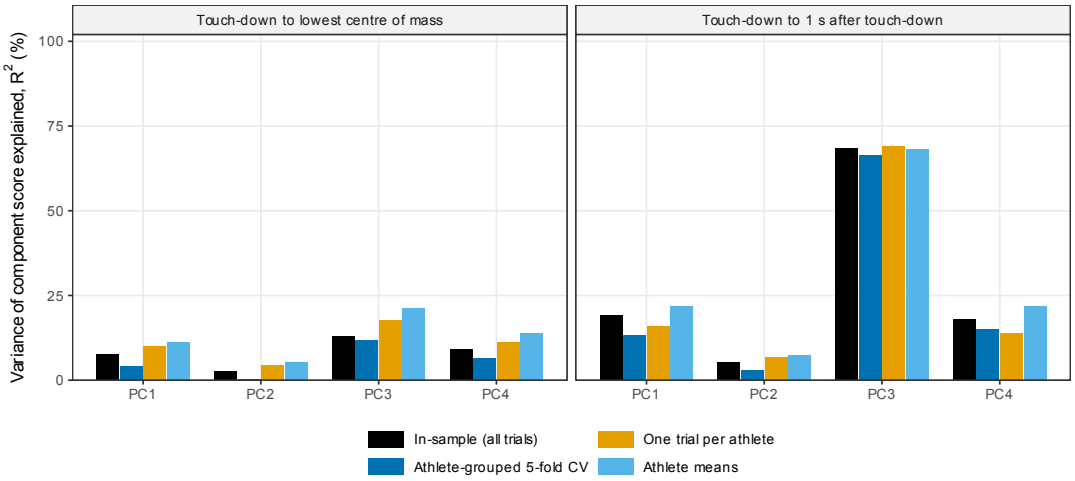


Figure 5. Proportion of the variance of each component score explained by the four vendor metrics, age and body mass, under four evaluation schemes, for the main landing definition (left) and the 1-s landing window (right). CV, cross-validation with athletes allocated to folds.

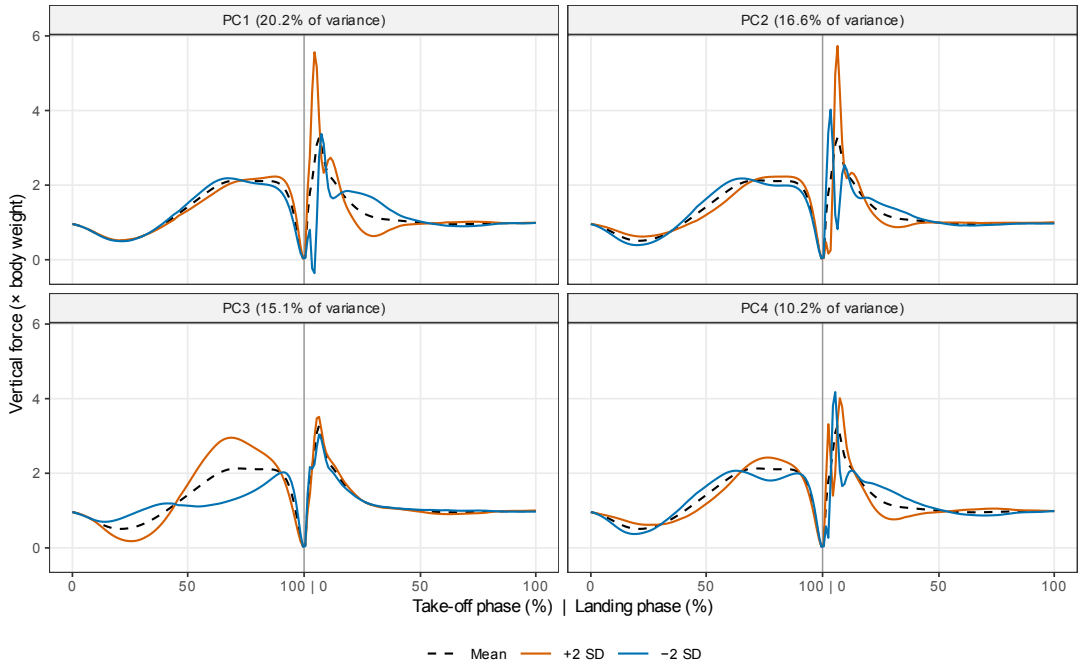


Figure 6. Sensitivity analysis with the landing phase ending 1 s after touch-down: mean curve (dashed) ± two standard deviations of each retained component (vermillion, +2 SD; blue, −2 SD). Under this definition PC3 is located almost entirely in the take-off phase (96.3% of its squared loading) and describes the duration and amplitude of the propulsive force.

Figure captions

Figure 1. Analysis pipeline. Steps 1–4 build one 202-point, body-weight-normalised force–time curve per trial; steps 5–7 address the three aims; step 8 examines the structure of the repeated measurements and the sensitivity of the conclusions to analytical choices. All decision rules were fixed before the results were examined. CMJ, countermovement jump; PCA, principal component analysis; OLS, ordinary least squares; CV, cross-validation; CI, confidence interval.

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Figure 4. Trials matched on jump height do not share a common force–time curve. (A) Curves of the 82 trials between the 45th and 55th percentiles of jump height (27.1 ± 0.65 cm), coloured by their PC1 score. (B) Ratio of the standard deviation of each component score among trials matched on jump height (circles) or on relative concentric peak force (triangles) to that in the full sample, with athlete-level bootstrap 95% confidence intervals. A ratio of 1 indicates that matching on the metric leaves curve-shape variation unchanged.

Figure 5. Proportion of the variance of each component score explained by the four vendor metrics, age and body mass, under four evaluation schemes, for the main landing definition (left) and the 1-s landing window (right). CV, cross-validation with athletes allocated to folds.

Figure 6. Sensitivity analysis with the landing phase ending 1 s after touch-down: mean curve (dashed) ± two standard deviations of each retained component (vermillion, +2 SD; blue, −2 SD). Under this definition PC3 is located almost entirely in the take-off phase (96.3% of its squared loading) and describes the duration and amplitude of the propulsive force.